

Epidemiological data from the [Integrated Disease Surveillance Programme \(IDSP\)](#) and the [Brihanmumbai Municipal Corporation \(BMC\)](#) from 2022 to 2026 reveals a clear structural framework of monthly seasonal disease patterns in India, specifically shifting over the last 5 years due to changing monsoon windows .

The annual calendar follows three core phases: **Winter Respiratory Loop** (Dec–Feb), **Summer Gastroenteric Wave** (Mar–May), and the **Monsoon/Post-Monsoon Hyper-Burden** (Jun–Nov).

5-Year Structural Trend (2022–2026)

Below is the master baseline mapping of seasonal vectors across Maharashtra.

Month-by-Month Detailed Epidemiological Analysis

Phase 1: Winter Respiratory Domain (December – February)

- **December:** Onset of cooler night temperatures drives viral stability. Steady upward rise in **Influenza A (H3N2)** and Acute Respiratory Infections (ARI).
- **January: Peak respiratory period.** High ambient humidity and temperature inversions concentrate urban pollutants, exacerbating Asthma, COPD, and RSV (Respiratory Syncytial Virus).
- **February:** Winter viral infections show a slow decay. Towards late February, chickenpox and measles outbreaks start ticking upward in rural pockets.

Phase 2: Summer Gastroenteric Shift (March – May)

- **March:** Transition month. As temperatures cross 35°C, water scarcity forces dependence on compromised alternate water lines. **Acute Diarrheal Disease (ADD)** cases begin climbing.
- **April: Peak Typhoid and Hepatitis A/E transmission.** Bacterial replication rates multiply in unhygienic local storage vats.
- **May: Food Poisoning & Gastroenteritis maxima.** High ambient heat accelerates food spoilage. In recent years (especially **2025**), the earliest monsoon onset in 75 years occurred in late May, which created a rare overlap where early flash rains flooded dry pipelines, causing hyper-localized gastroenteritis clusters.

Phase 3: Monsoon & Post-Monsoon Hyper-Burden (June – November)

- **June: The Flood Surge.** Early seasonal monsoons trigger waterlogging. High spikes in **Leptospirosis** (due to wading in rodent-urine contaminated water) and initial **Malaria** vector

breeding in fresh puddle formations.

- **July:** Heavy continuous downpours flush out open larvae but result in household water contamination. **Cholera, Dysentery, and Gastroenteritis cases spike** drastically.
- **August: The Vector Explosion.** Intermittent rainfall replaces continuous downpours, leading to thousands of artificial puddles. **Malaria and Dengue transmission curves accelerate exponentially.**
- **September: The Critical Absolute Peak.** This month marks the worst public health burden of the year. Urban construction zones, high rises, and scrap yards act as massive amplification vectors for *Aedes aegypti* (Dengue) and *Anopheles* (Malaria) mosquitoes. Concurrently, high humidity brings the **secondary bimodal Influenza/H1N1 peak.**
- **October: The Post-Monsoon Tail.** Stagnant receding waters allow continuous breeding. Dengue hemorrhagic presentations typically peak in clinical admissions during this dry-hot post-monsoon phase. **Scrub Typhus** cases register heavily in rural and semi-urban perimeters.
- **November:** Onset of cold winds diminishes vector populations. Cases fall by 50–60% compared to September metrics, easing hospital bed occupancy.

Data Evolution: 2022 vs. 2026 Shift

The shift over the last 5 years indicates that while waterborne bacterial outbreaks are stabilizing due to better civic infrastructure, **vector-borne and viral respiratory diseases are escalating dramatically.**

Disease Metric (Monsoon Window)	2022–2023	2025 Records	2026	5-Year Vector Shift Vector
	Baseline Average		Recent Records	
Malaria Cases (Mumbai Core)	~3,500	4,304	5,984	 Sharp upward linear volume.
H1N1 (Swine Flu) Cases	~75	83	488	 ~600% explosion due to bimodal humidity shifts.
Gastroenteritis Cases	~6,500	5,510	3,866	 Gradual drop due to pipeline sanitation.
Leptospirosis Tracked	~110	136	157	 Creeping rise due to intensive

This structural shift over the 2022–2026 period underscores a key reality for public health deployment: **The traditional monsoon risk window has expanded.** Disease management protocols that previously targetted June–August must now run intensively from **May through late November** to match the realigned climate patterns.